TAP · PAY · RECORD

Hydra Pay

Tap a card. Get an M-Pesa prompt.
Every shilling confirmed and recorded.



HYDRA-PAY · Technical University of Kenya

DTMES 2026 · Africa Electronics Innovation Challenge

THE PROBLEM

Busy counters can't stop to check every payment

99%

of Kenyan small businesses still run partly or fully on cash

86%

already accept digital payments, so they juggle both

Fake SMS

Screenshots and forwarded texts get accepted as "proof" during a rush

Paper books

Sales are written by hand, so closing the day means guessing

Matatus, canteens and market stalls serve a queue in minutes. **Speed wins, and checking payments loses.**

Source: FinAccess MSE Tracker Survey, 2023.

THE SOLUTION

One tap starts the payment. Safaricom confirms it.

01

Tap

The customer taps their Hydra Pay card on the reader. The LED blinks and the tap is logged.

→

02

Pay

An M-Pesa prompt for the card's price pops up on the owner's phone. They enter their PIN. Nothing to type.

→

03

Record

Safaricom confirms. The dashboard shows **PAID** with the M-Pesa code and a WhatsApp receipt goes out.

Paid means paid. Status comes straight from Safaricom, never from a screenshot or an SMS the customer shows you.

THE HARDWARE

ESP32 + RC522: a reader from two low-cost parts

RC522 → ESP32 (SPI)

RC522	ESP32
SDA (SS)	GPIO 5
SCK	GPIO 18
MOSI	GPIO 23
MISO	GPIO 19
RST	GPIO 22
3.3V	3V3 (not 5V)
GND	GND

- **13.56 MHz RFID** (MIFARE cards and key tags) read over SPI at 1 MHz
- **MicroPython firmware** checks the RC522 firmware version at boot and retries until the wiring is good
- **Debounce:** holding a card on the reader counts as one tap, not ten (1.5 s window)
- **Feedback:** on-board LED flashes on every accepted tap
- **Powered at 3.3 V** because 5 V can damage the RC522
- Sends `UID:XX XX XX XX` to the payment server over USB serial

HOW IT RUNS

From card tap to M-Pesa receipt

READER	ESP32 + RC522 reads the card UID and sends it to the Hydra Pay server.
PAYMENTS	Python server looks up the card owner and price, then sends an M-Pesa STK push through Safaricom's Daraja API.
VERIFICATION	Safaricom's callback marks the payment PAID with its M-Pesa code. If the callback is late, the server polls Daraja for up to 3 minutes.
RECORDS	Every tap and payment is saved in a SQLite database: card, amount, phone, status, M-Pesa code and time.
DASHBOARD	Live web dashboard to register cards, set prices, watch payments and export records. Admin actions need a PIN.
CUSTOMERS	WhatsApp receipts and a menu (my card, my payments), plus USSD for phones without internet.

ENGINEERING FOR TRUST

Built so a payment is never marked wrong

VERIFIED, NOT CLAIMED

Only Safaricom's result can set a payment to PAID. Cancelled or unanswered prompts are recorded as NOT PAID with the reason.

TWO PATHS TO THE ANSWER

Callback first, polling as backup. "Still processing" is treated as pending, not as failed.

RECORDED EXACTLY ONCE

A lock ensures each payment is finalised once, even when the callback and polling answer together.

NO DOUBLE CHARGING

A card can't get a second prompt while one is still waiting on the owner's phone.

PROOF OF WORK

Built, wired and working end to end

27

M-Pesa prompts sent from real card taps during testing

10

payments confirmed PAID by Safaricom, each with its M-Pesa code

15

cancelled or unanswered prompts correctly recorded as NOT PAID

Two modes: **pay per tap** (M-Pesa prompt) or **prepaid wallet** (cash or M-Pesa top-ups)

Tested on Safaricom's **Daraja sandbox**. Live demo at our table.

WHO IT'S FOR

Small businesses where every second counts

START HERE

Canteens

Lunch-hour queues, fixed prices and regular customers. A tap is faster than typing a till number.

NEXT

Matatus

Tap IN when boarding, tap OUT when alighting. The conductor sees every confirmed fare.

THEN

Market traders

Mixed cash and M-Pesa, with one record of the day's sales instead of a notebook.

Our first pilot: **canteens at the Technical University of Kenya**, where we can measure checkout time before and after.

WHAT IS NEW

Tap like a bank card.
Pay like M-Pesa.

No typing

No till number, no amount, no account to enter. Tap, then PIN.

No fake proof

The seller never has to trust an SMS. Safaricom's answer is on the dashboard.

Any phone

Receipts on WhatsApp, balance and payments by USSD. No app to install.

GROWTH

Three steps to the street

NOW

Working prototype

Reader, M-Pesa prompts, dashboard, WhatsApp and USSD running end to end on the Daraja sandbox.

NEXT

Campus pilot

Live Safaricom shortcode, a cased reader on its own Wi-Fi with no laptop, and canteen trials at TU-K.

THEN

Matatus and markets

Battery-powered reader that queues taps when the network drops, then expand to transport and traders.

Revenue: a small monthly fee per reader. No cut of each sale, so low-value, high-volume sellers keep their margin.

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